

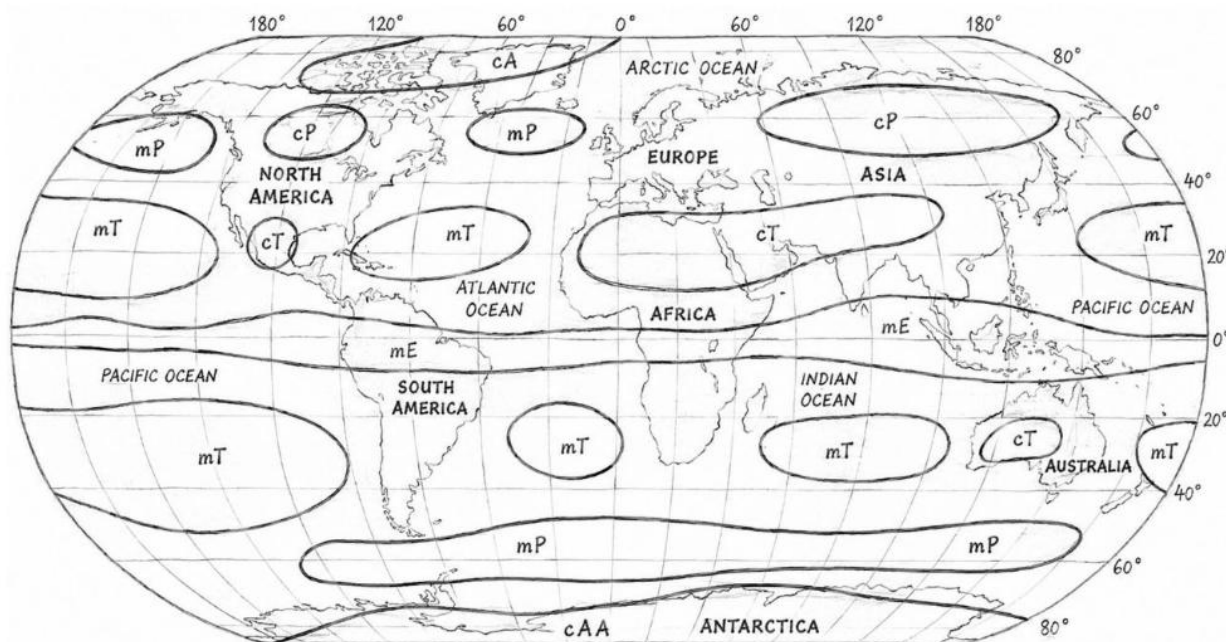
GALLANT IAS ACADEMY
PRELIMS CUM MAINS BATCH 2027

GEOGRAPHY
CLIMATOLOGY

MODULE 6

AIR MASSES, CYCLONES AND TORNADOES

Air masses are classified at two levels: **Geographical (primary)** and **Thermodynamic and Mechanical (secondary)**.



Geographical Classification (Primary)

According to **Trewartha**, air masses can be classified based on the properties of their source regions. Based on the latitude of source regions: **Polar air mass (P)** - which also includes Arctic (A) and Antarctic (AA); and **Tropical air mass (T)** - which also includes Equatorial air mass (E). Based on the nature of the source region (land or ocean): **Maritime air mass (m)** - originally having large quantities of moisture picked up from oceans; and **Continental air mass (c)** - which are generally dry. This gives rise to **four principal types**:

Type	Code	Character
Continental polar	cP	Extremely cold and dry; originates over central Canada and Siberia.
Maritime polar	mP	Cold and moist; forms when cP moves over warmer oceanic surfaces.
Continental tropical	cT	Very high temperature, low moisture, dry; originates in subtropical hot deserts near 30°.
Maritime tropical	mT	Warm, moist and unstable; source regions over warm ocean surfaces between 30°N and 30°S.

Thermodynamic and Mechanical Classification (Secondary)

This classifies air masses based on their modification after leaving their source regions. Two types of modifications occur:

Thermodynamic modification refers to change in temperature properties. **Cold air masses (K):** temperature of the air mass is colder than the surface it is passing over; generally unstable. **Warm air masses (W):** temperature of the air mass is higher than the surface; more stable; temperature inversion common.

Mechanical modification occurs when air enters zones of convergence (low pressure belts - air forced to ascend, becomes unstable - depicted by 'u') or zones of divergence (high pressure belts - air made more stable - depicted by 's'). Combining thermodynamic (K/W) and mechanical (s/u) modifications with the four principal types gives **16 sub-types** (e.g., cPKs, mTWu).

CHARACTERISTICS OF MAJOR AIR MASSES

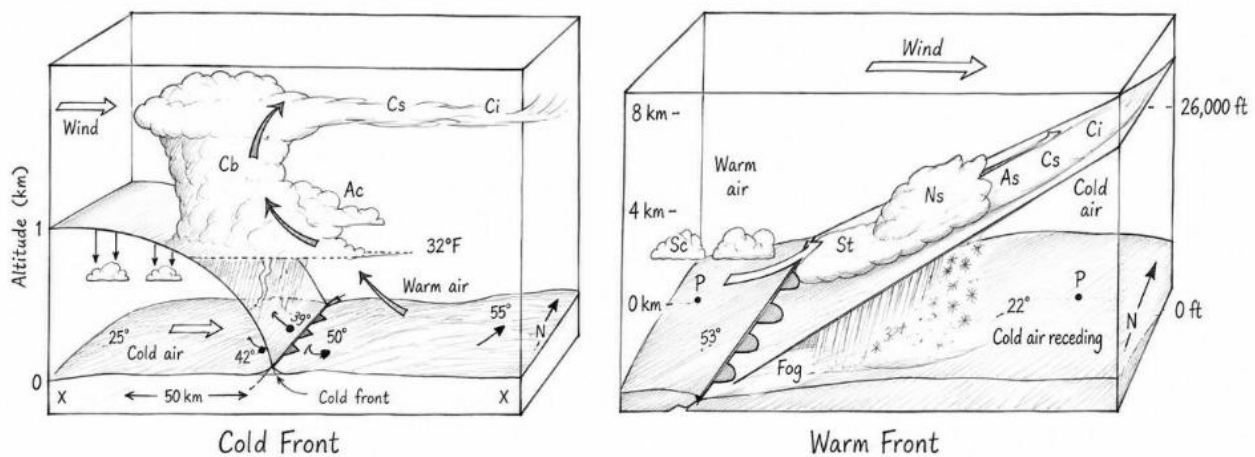
Air Mass	Source Region	Properties	Key Examples
Continental Polar (cP)	Central Canada, Siberia, Arctic basin, Greenland	Extremely cold and dry. Intense cold waves in areas they visit. When they move over warmer surfaces, they are heated from below and become unstable.	Lake-effect snow: cP from Canada modified over Great Lakes becomes moist and yields heavy snowfall. Arctic smoke: very cold/dry cA from Arctic moving over the warm Norwegian Sea becomes unstable, forming vertical trails of condensation.
Maritime Polar (mP)	Same as cP but modified over oceanic surfaces of higher latitudes	Cold and moist. When cP moves over the ocean, its lower parts are heated from below and become mP.	On the west coast of North America, mP forces precipitation on windward (Coast Ranges) while leeward side (Rockies, Great Plains) goes dry due to adiabatic compression on descent.
Continental Tropical (cT)	Subtropical high-pressure belt of hot deserts near 30° latitudes	Very high temperature, very low moisture content, dry weather. Rarely moves out of its source region.	When cT moves over ocean surfaces, it is modified into maritime tropical air mass.
Maritime Tropical (mT)	Warm ocean surfaces of tropical regions between 30°N and 30°S	Warm, moist and unstable; more extensive in area. Produces cumulus and cumulonimbus clouds. Gives abundant rainfall over mountains or across fronts.	When mT moves towards higher latitudes, cooler surfaces stabilize it.

FRONTS

A front is a **zone of transition formed in the middle latitudes between two contrasting air masses** (differing in temperature, humidity and density). Also called a **frontal zone or frontal surface**. It is a sloping boundary - inclined at a low angle, neither parallel nor vertical to the ground surface. Fronts are less significant in higher latitudes and rare in the tropics because air masses in those regions tend to be similar. The process of formation of fronts is called **frontogenesis** (caused by convergence of contrasting air masses). The destruction of a front is called **frontolysis**.

Pre-conditions for front formation: (1) Presence of two drastically different air masses differing in temperature and humidity; and (2) Atmospheric circulation causes them to converge forcefully. These conditions are met at mid-latitudes (40°-60° N or S), where westerlies and polar easterlies bring polar and tropical air masses together. Frontogenesis is therefore a **phenomenon of mid-latitudes**.

Types of Fronts

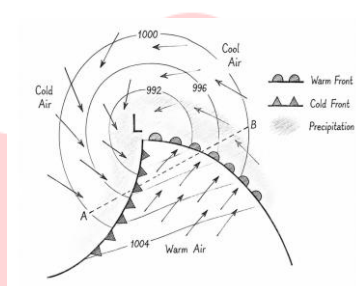


Type	Mechanism	Slope	Weather	Post-passage Conditions
Cold Front	Cold air mass is aggressive and actively undercuts warm air mass, forcing it to rise steeply.	Steeper (1:50 to 1:100)	Heavy precipitation of short duration, cumulonimbus clouds, thunderstorms, hailstorms. Small areas affected. The weather is intense. Called 'squall lines' in USA.	Clouds disappear; decrease in temperature and humidity; increase in air pressure; clear weather.
Warm Front	Warm and lighter air is aggressive and rises slowly and gently over cold and dense air.	Gentle (1:100 to 1:400)	Gradual upliftment causes moderate to gentle precipitation over a large area for longer duration	Warm sector arrives - sudden increase in temperature and humidity, decrease in air pressure, clear

			(several hours). Mostly nimbostratus. Similar to orographic precipitation.	sky, break in precipitation.
Stationary Front	Two contrasting air masses converge parallel to each other with no ascent of air. The surface position does not move forward or backward.		Not associated with cloud formation or precipitation.	No major change.

Occluded Front

Formed due to the phenomenon of occlusion. Occluded front is formed when cold front (faster) overtakes warm front and warm air is completely lifted from the ground surface. This leads to extensive cloudiness and rainfall which can be light to heavy depending upon the nature of warm air. The weather conditions under such fronts are complex with rapidly changing winds and precipitation types.



Frontal Fogs

Fog produced along warm and cold fronts forms due to the **cooling of overlying warm moist air by the underlying cold air**. Warm air being lighter is pushed upward by the heavier cold air and is chilled from below, resulting in saturation and condensation of the warm air. Such fogs are mostly formed in the middle latitudes.

Classification of Winds

Winds can be divided into:

Primary Circulation / General circulation of the atmosphere- which includes all the planetary wind systems.

Secondary Circulations- It consists of cyclones, anticyclones, monsoons, and air masses.

Tertiary Circulations- It includes the local winds. The secondary and tertiary circulations are variations that take place over and above the primary circulations. They are also called atmospheric disturbances.

CYCLONES

A cyclone is a **wind system with low pressure at the centre**, surrounded by closed isobars with increasing pressure outwards. They are rotating systems - as air enters the low-pressure region, the Coriolis effect deflects it. Winds in a cyclone move **anticlockwise in the northern hemisphere and clockwise in the southern hemisphere**. A cyclone is a system of low-level convergence and high-level divergence with a rising column of air at the centre. The winds spiral and rise rapidly along the core, resulting in cloud formation and heavy rainfall - called **cyclonic precipitation**.

Based on location, cyclones are broadly classified into **tropical cyclones** (thermally induced; found in tropical regions; embedded in trade winds - move east to west) and **temperate cyclones** (dynamically induced; found in mid-latitudes; embedded in westerlies - move west to east).

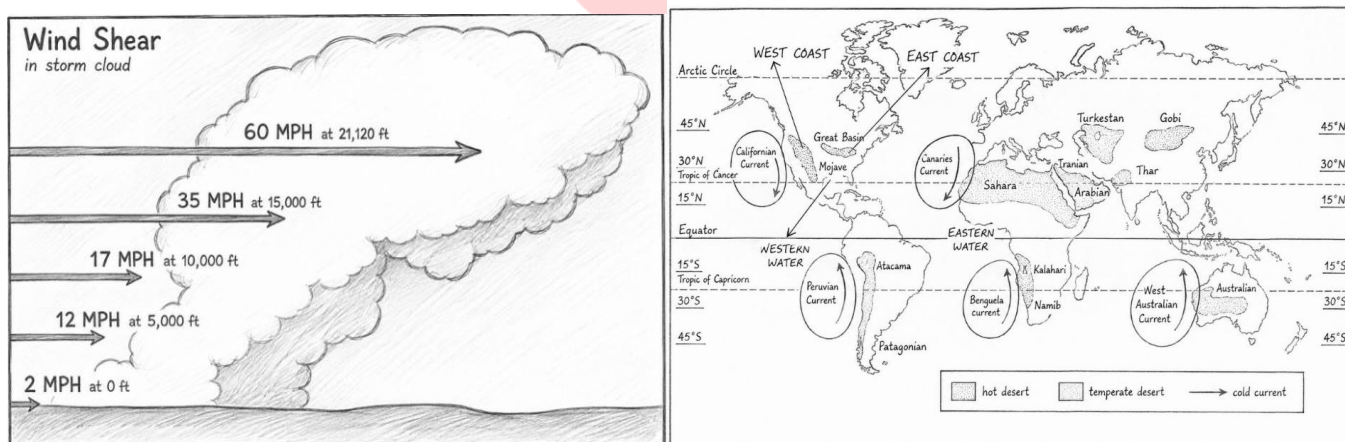
TROPICAL CYCLONES

Also called **warm core cyclones** since they are thermally induced. They are intense low-pressure systems created over the tropical regions and are among the most violent

atmospheric disturbances. They usually develop during the summer season in the vicinity of the ITCZ. Key conditions for development:

- **Large and continuous supply of moisture** - they always develop over water bodies, die when they hit land. The point where a cyclone crosses the coast is called the **landfall of the cyclone**.
- **Strong Coriolis force** - ITCZ or low-pressure forms at 5°-8° away from the equator. Most tropical cyclones are created between 5° to 20° latitudes.
- **Sea Surface Temperature (SST) is more than 27°C.**
- **Weak vertical wind shear** - a strong high-pressure in the upper troposphere coupled to surface low-pressure. Strong wind shear can decouple aloft divergence from surface convergence, preventing tropical cyclone formation. Tropical cyclones are usually not created during the southwest monsoon because of strong vertical wind shear.
- **Created on tropical western waters** - they do not develop on tropical eastern water (west coast) which has cold currents and upwelling of cold water.

Salient Features of Tropical Cyclones



Feature	Detail
Pressure	Acute low pressure at the centre (as low as 960 mb). Central pressure 50-60 mb lower than the outer edge. Lowest ever: 870 mb at Super Typhoon Tip (1979).
Wind Speed (WMO)	119 km/hr minimum for tropical cyclone; 63-118 km/hr = tropical storm; <63 km/hr = tropical depression.
Wind Speed (IMD)	63 km/hr minimum for cyclonic storm; 31-50 km/hr = depression; 51-62 km/hr = deep depression; 221+ km/hr = super cyclonic storm.
Frequency	Less frequent than temperate cyclones but much more destructive. Mainly formed in the summer season.
Movement	East to west under trade winds at 300-500 km/day. Sometimes stationary for several days.
Diameter	About 150-250 km; can extend up to 600 km or more.
Life span	Usually 4-5 days. Weakens after landfall or when moving into cooler waters.

Structure of a Tropical Cyclone

Part	Description
Eye	Central region; diameter 16-24 km. Lowest atmospheric pressure. Low wind speed, high temperature, high relative humidity, almost clear sky. Characteristic features of tropical cyclones (not developed in temperate cyclones).
Eyewall	Narrow band of highest velocity winds surrounding the rim of the eye. Ring of cumulonimbus clouds around the eye.
Inner Cyclone	Region of rising air (updraft). Heavy rainfall along with hailstorms.
Annular Belt / Downdraft Region	Region where air sinks. Rainfall recedes or weakens.
Outer / Peripheral Region	Beyond the sinking region. Receives rainfall from rainbands (spiral/feeder bands) - branches of cumulonimbus clouds radiating into the outer region.

Stages of Formation of Tropical Cyclones

Stage	Description
Formation and Initial Development	Low-pressure system developing; increasing wind speeds; cumulonimbus not yet fully developed; eye and eyewall not yet formed. Low pressure sustained by latent heat of condensation released by moisture condensing over the warm ocean.
Mature Stage	Central low pressure strengthens; steep pressure gradient; high velocity winds; eye and eyewall well-formed; fully developed cumulonimbus; intense rainfall; spiralling rain bands extend outward.
Modification and Decay	Cyclones weaken once the source of warm moist air is cut off - after landfall or on passing over cold waters. Central low pressure, internal warmth and high wind speeds all decline.

Regional Names of Tropical Cyclones

Regional Name	Region
Hurricane	Northern Atlantic and Eastern North-Pacific
Typhoon	Western North-Pacific
Baguio	Philippines
Taifu / Nowaki	Japan
Willy-Willies	Australia
Cyclone	Indian Ocean Region

Cyclonic Destruction

Cyclones cause huge destruction in coastal areas due to high wind velocity and floods. Floods are caused by: (1) cumulonimbus clouds yielding heavy precipitation; and (2) **Storm surge / Tidal surge** - rise of seawater caused by strong low pressure and strong winds as the cyclone approaches the coast. The **1999 Odisha super cyclone** created a storm surge of 9m, transgressing up to 20 km inside the coastal region and resulting in the death of 10,000 people (unofficial sources put the death toll at 1 lakh). The **1970 cyclone of Bangladesh** (and parts of West Bengal) killed more than 3 lakh people. Cyclone forecasting, tracking and better disaster management have reduced the death toll in recent years despite the increasing frequency of tropical cyclones due to climate change.

India Meteorological Department
Tropical Cyclone Intensity Scale

Category	Sustained winds (3-min average)
Super Cyclonic Storm	≥120 kt ≥221 km/h
Extremely Severe Cyclonic Storm	90–119 kt 166–220 km/h
Very Severe Cyclonic Storm	64–89 kt 118–165 km/h
Severe Cyclonic Storm	48–63 kt 89–117 km/h
Cyclonic Storm	34–47 kt 63–88 km/h
Deep Depression	28–33 kt 51–62 km/h
Depression	17–27 kt 31–50 km/h

Relative Absence in South Atlantic and Southeast Pacific

Tropical cyclones are relatively absent in the South Atlantic and Southeastern Pacific because:

- (1) the ITCZ seldom shifts far enough into these locations - tropical cyclones require low pressure at least 5°-8° from the equator; and
- (2) the water is not warm enough in these locations.

Tropical Cyclones in India

Many low-pressure systems develop in the Bay of Bengal and the Arabian Sea. Most form in the Bay of Bengal, due to higher sea surface temperatures. The systems reach maximum intensity **before or after the southwest monsoon season** - from April to June and from October to December. Tropical cyclones are more frequent and intense during the **retreating monsoon season** (October onwards) because of low vertical wind shear and warmer waters in the late summer.

Most tropical cyclones are created between 5° to 20° latitudes and move westward under the influence of trade winds. Those that cross 20°N generally **recurve towards north or northeast** and tend to be more destructive. Even after reaching mid-latitudes, if not exhausted, they move eastward under the influence of westerlies.

Fujiwhara Effect

The **Fujiwhara Effect** is a unique meteorological phenomenon that occurs when two tropical cyclones spinning in the same direction come within **1,400 kilometres** of each other. Instead of moving independently along their normal paths, the two storms begin an interactive orbital movement around a shared centre of low pressure

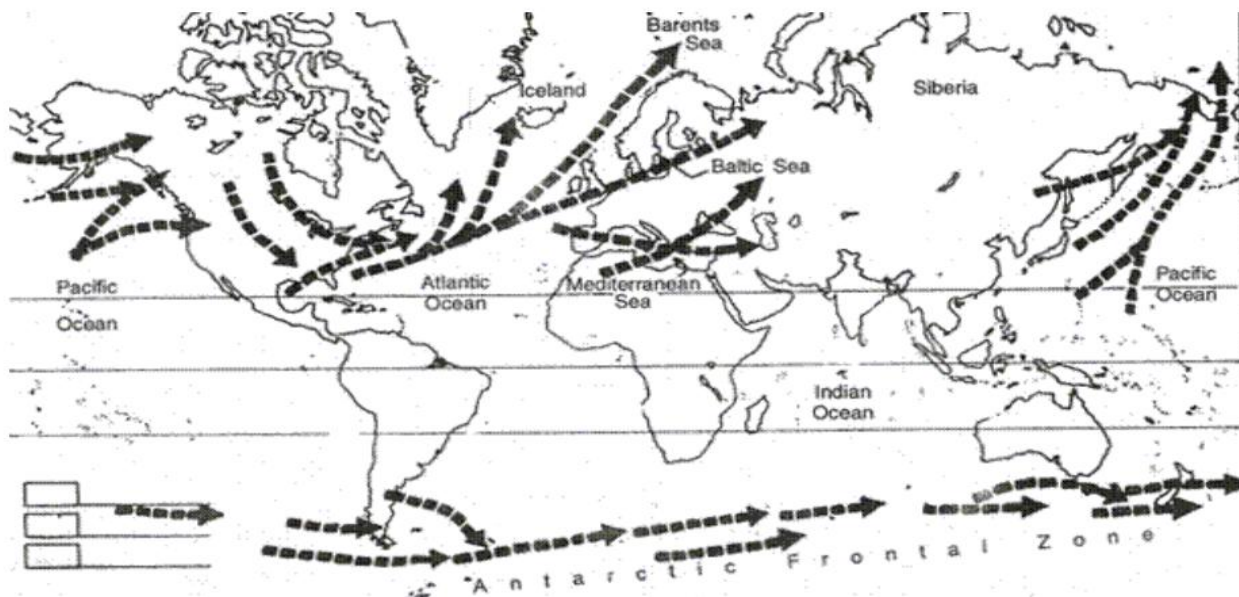
Types of Interactions and Outcomes

The ultimate outcome of this interaction depends entirely on the size, intensity, and distance between the two systems. If one cyclone is significantly larger and more dominant, it can completely absorb or tear apart the smaller system, creating a singular, hyper-intense mega-storm. Conversely, if the two cyclones are similar in size, they can experience an **elastic interaction**, where they warp each other's paths and alter track directions before safely separating.

Disaster Management Challenges

Fujiwhara Effect presents a severe challenge because it makes storm tracks highly unpredictable. Standard weather forecasting models struggle to map the erratic movements, loops, and sudden stalls caused by the binary interaction. This shrinkage of early warning timelines directly complicates evacuation protocols and exponentially increases the risk of prolonged rainfall, severe flooding, and coastal storm surges.

TEMPERATE CYCLONES



Also called **extratropical / cold core / frontal / wave / travelling / mid-latitude cyclones**. They are one of the most significant atmospheric disturbances in the mid and higher latitudes. These systems are **frontal in origin** - they receive their energy from the interaction of warm and cold air masses. The polar front, created by the interaction of cold polar and warm tropical air masses, is responsible for their development. Their development is also determined by upper atmospheric circulations.

Polar Front Theory by Bjerknes

Also known as '**frontal theory**', '**wave theory**' or '**Bergen theory**', propounded by **Vilhelm Bjerknes and his son Jakob Bjerknes**, two famous Norwegian meteorologists working at Bergen, Norway. According to this theory, mid-latitude cyclones are created due to the collision of two air masses of contrasting characteristics - cold polar air masses meet warm tropical air masses in the temperate zone around 60° latitude. They do not merge readily but form a **Polar Front**. The theory suggested four stages of formation.

Stage	Description
1. Initial Stage	A stationary front is created between the cold polar air mass and the warm tropical air mass. Cold polar air blows eastward poleward of the front; warm tropical air blows westward equatorward. Neither tries to encroach into the other's area.
2. Incipient Stage	A low pressure forms along the stationary front (best explained by baroclinic theory). Winds begin to blow anticlockwise (NH) around the low. West of the low: cold air moving from north creates a cold front. East of the low: warm air moving from the south creates a warm front.
3. Mature Stage	Fronts are fully developed. A cold front (faster) narrows the warm sector. Warm air actively rises along fronts producing clouds and precipitation. Wind velocity increases to ~60 km/hr. Heavy rainfall especially along the cold front with thunderstorms. Isobars are fully elliptical.
4. Occlusion Stage	Faster cold front catches up with the warm front, fully lifting the warm air mass and creating an occluded front. Extensive cloudiness and rainfall. As cold air masses mix and lose

	temperature contrast, fronts and low pressure disappear - the cyclone dies.
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The formation of low pressure in temperate cyclones is not fully understood. The **baroclinic theory** offers the best explanation through: (1) upper tropospheric meandering jet streams (Rossby waves) inducing pressure conditions; (2) temperature contrast between air masses inducing low pressure at the front; and (3) a weak tropical cyclone being carried into the middle latitudes by Rossby waves.

Tropical vs Temperate Cyclones

Feature	Tropical Cyclones	Temperate Cyclones
Origin	Thermal (heat-induced over warm oceans)	Dynamic (baroclinic - frontal; interaction of air masses)
Front	Not associated with fronts	Frontal in origin (polar front)
Temperature variation	Mostly warm throughout; no temperature variation in different parts	Temperature varies in different parts due to cold and warm air masses separated by fronts
Eye	Well-developed eye characteristic feature	No eye development
Size	Smaller: 150-600 km diameter	Larger: 800-1600 km diameter
Pressure gradient	Very steep; high wind velocity	Weak; average 30-50 km/hr wind velocity
Source region	Only over warm ocean surfaces (>27°C SST)	Can originate over land and water
Movement	East to west (trade winds)	West to east (westerlies)
Season	Mainly summer season	Can occur throughout the year; more intense in winter
Frequency	Less frequent	More frequent
Destructiveness	Much more destructive	Less destructive
Life span	4-5 days	10-15 days
Isobars	Mostly circular	Egg-shaped (elliptical)
Rainfall	Cyclonic precipitation (intense, violent)	Frontal precipitation; moderate to heavy along fronts; responsible for uniform rainfall in mid-latitudes

Western Disturbances

Western disturbances are **weak temperate cyclones occurring over northern India during winter (October to February)**, captured by the subtropical westerly jet streams. They originate in West Asia, the Mediterranean region and even the Atlantic Ocean and are carried eastwards. Moisture content is increased as they move over the Red Sea and Persian Gulf. They are very weak by the time they reach the subcontinent. On reaching Rajasthan, Punjab and Haryana, these disturbances slow down due to the nearly closed-in nature of the region with high hills; with moisture fed from the Arabian Sea, they may intensify.

Effects: (1) **Light rainfall** in the northwestern plains - beneficial for **rabi crops** like wheat; (2) **Snowfall** in the higher altitudes of Jammu and Kashmir, Himachal Pradesh; (3) **Cold waves** in the region; (4) **Hail formation** at times.

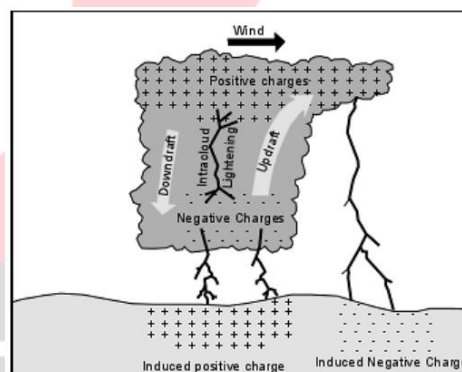
ANTICYCLONES

Anticyclones are **high-pressure systems where the air descends**. They rotate **clockwise in the northern hemisphere and anticlockwise in the southern hemisphere**. The highest pressure is recorded at the centre and pressure decreases outward. Anticyclones have **low wind speeds due to a weak pressure gradient** - the pressure difference between centre and periphery ranges between 10-20 millibar; average wind velocity of 30-50 km/hr. They are much larger than temperate cyclones - sometimes close to 9,000 km in diameter. Since air is continuously descending, anticyclones are associated with **calm, clear weather and rainless conditions**.

Type	Origin	Characteristics	Example
Cold Anticyclones	Higher latitudes (Arctic regions)	Cold high-density air of higher latitudes creates high pressure, especially in winter.	Cold air from northern Canada's winter anticyclone brings cold waves to Canada and the USA. The Siberian anticyclone affects China, Japan and Alaska.
Warm Anticyclones	Subtropical high-pressure belt	Warm, subsiding air; stable, dry, clear weather.	Found near 30°N and 30°S - the subtropical high-pressure belt or 'Horse Latitudes.'

THUNDERSTORMS

A thunderstorm is a **well-grown cumulonimbus cloud producing heavy rainfall with thunder and lightning**, characterised by strong updrafts of air. Instability can be created by intense heating of the surface, topography or the presence of cyclones. They are of **short duration and cover a small area** (localised) but are violent in nature and can produce hailstorms. Thunderstorms occur over land and water but are more frequent over landmasses since land warms more rapidly to produce strong convection.



Stages of Thunderstorm Development

Stage	Key Process
Cumulus Stage	Surface heating creates updrafts forming fair-weather cumulus clouds, growing vertically into cumulonimbus. Bergeron process begins above freezing levels producing precipitation. Falling precipitation creates drag initiating downdrafts.
Mature Stage	Both updrafts and downdrafts exist side by side. Heavy rainfall from cumulonimbus accompanied with maximum thunder and lightning. Full development of cumulonimbus clouds. Most active period.
Dissipating Stage	Downdrafts begin to dominate. Cooling effects of precipitation end thunderstorm activity by creating atmospheric stability. Without continuous moisture from updrafts, clouds soon dissipate.

Classification of Thunderstorms

Type	Mechanism	Key Features
Thermal / Heat / Local	Intense surface heating creates convection currents.	Mostly in the afternoon, summer season; short duration; influenced by limited areas; more frequent in lower latitudes.
Orographic	Warm, moist air forced to rise hurriedly along a mountain slope.	More extensive than thermal thunderstorms.
Frontal	Especially associated with the cold front in a temperate cyclone.	Warm front thunderstorms also exist but are weak and not significant.

Thunderstorm Electricity and Thunder

In the mature stage, **positive charges accumulate in the upper portions and negative charges in the lower portion** of the cloud. When the electrical potential gradient becomes very steep, **lightning** discharges occur - within a cloud, between cloud and earth, or between clouds. Lightning produces tremendous heat along its path, heating the air to **10,000°C**, causing rapid expansion of air. The sound emitted by such rapidly expanding gases is called **thunder**, which can be heard up to **15 to 20 km**.

Characteristics	Thunderstorms	Tropical Cyclones
Nature of air movement	Non-rotating systems	Rotating systems
Duration	Short lived for few hours	Relatively longer life span (4-5 days)
Area affected	Small	Large
Wind speed	Relatively low wind speeds	High wind speeds
Location of formation	Land or water (mostly land)	Warm ocean surfaces

CLOUDBURSTS

A cloudburst is a term used for **sudden, intense and highly concentrated rainfall** over a generally small area, mostly of shorter duration. Usually, rainfall is greater than **10 cm per hour**. They are mainly the result of thunderstorms - created by thermal convective thunderstorms and orographic thunderstorms. In India, cloudbursts frequently occur in the Himalayan regions (Uttarakhand, Himachal Pradesh) because tall mountains force moist monsoon air to rise quickly, and are also reported from parts of the Western Ghats.

Major Cloudburst Incidents in India

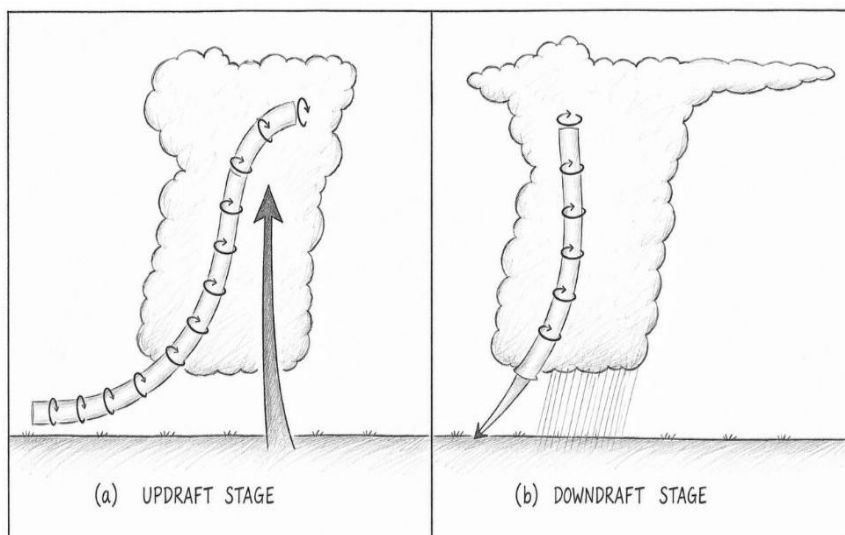
Incident	Impact
Leh, Ladakh (2010)	Massive floods and landslides; extensive damage and numerous fatalities.
Uttarakhand (Kedarnath) (2013)	Catastrophic floods and landslides; more than 6,000 people died.
Amarnath, Jammu & Kashmir (2022)	Flash floods near Amarnath pilgrimage route; pilgrims and infrastructure affected.
Himachal Pradesh (2023)	Multiple cloudbursts causing flash floods, landslides, loss of lives and infrastructure destruction.

Impacts and Mitigation

Impacts: Flash floods (can trigger GLOF); landslides (soil saturation in vulnerable regions); infrastructure damage; loss of life and livelihood; ecosystem disruption affecting biodiversity and river systems.

Mitigation: Early warning systems; resilient urban planning with effective stormwater drainage; public education on hazards; watershed management; reforestation to absorb excess water and stabilise slopes; international cooperation for shared river basins.

TORNADOES



From severe thunderstorms (cumulonimbus clouds), spiralling winds descend like a **funnel-shaped storm** with very low pressure at the centre (even 100 mb less than the outside environment). Because of this steep pressure gradient, winds rush towards the centre at velocities of **more than 400 km/hr**. The diameter varies between **150 to 600 m**. They are **the smallest but most violent of all storms**. The tornado over the sea is called a **waterspout**. In the USA, tornadoes are referred to as

"twisters".

Tornadoes are usually associated with **super-celled thunderstorms / Mesocyclones** formed where a strong vertical wind shear produces a rolling motion about a horizontal axis. The maximum distribution is in the **Tornado Alley of USA** (part of the Great Plains of central United States), where the maritime humid air mass from the Gulf of Mexico collides with the continental polar air mass from Arctic Canada. Tornadoes can also occur in tropical regions (reported from Australia) and may develop individually or in groups called a **tornado family**.

PREVIOUS YEAR QUESTIONS [PRELIMS]

1. Consider the following statements about the 'Roaring Forties': [2000]
 - I. They blow uninterrupted in the Northern and Southern Hemispheres.
 - II. They blow with great strength and constancy.
 - III. Their direction is generally from North-West to East in the Southern Hemisphere.
 - IV. Overcast skies, rain and raw weather are generally associated with them.Which of these statements are correct?
 - (a) I, II and III
 - (b) II, III and IV
 - (c) I and IV
 - (d) None
2. Thunder Cloud development is due to which of these phenomena? [2002]
 1. Condensation
 2. High temperature and humidity
 3. Orography
 4. Vertical windWhich of these statements are correct?
 - (a) 1 and 2
 - (b) 2, 3 and 4
 - (c) 1, 3 and 4
 - (d) 1, 2, 3 and 4

3. Consider the following [2002]
Assertion (A): The surface winds spiral inwards upon the centre of the cyclone.
Reason (R): Air descends in the centre of the cyclone
Which of these statements are correct?
(a) Both (A) and (R) are individually true and (R) is the correct explanation of (A)
(b) Both (A) and (R) are individually true but (R) is not the correct explanation of (A)
(c) (A) is true but (R) is false
(d) (A) is false but (R) is true
4. Consider the following [2005]
Assertion (A): Wind patterns are clockwise in the northern hemisphere and anticlockwise in the southern hemisphere.
Reason (R): The directions of wind patterns in the northern and the southern hemisphere are governed by the Coriolis effect.
Which of these statements are correct?
(a) Both (A) and (R) are individually true and (R) is the correct explanation of (A)
(b) Both (A) and (R) are individually true but (R) is not the correct explanation of (A)
(c) (A) is true but (R) is false
(d) (A) is false but (R) is true
5. Consider the following statements: [2007]
1. Either of the two belts over the oceans at about 30° to 35° N and S Latitudes is known as Horse Latitude.
2. Horse latitudes are low pressure belts.
Which of the statements given above is/are correct?
(a) 1 only
(b) 2 only
(c) Both 1 and 2
(d) Neither 1 nor 2
6. Which of the statements given above is/are correct? [2009]
1. In the world, the tropical deserts occur along the western margins of continents within the trade wind belt.
2. In India, the East Himalayan region gets high rainfall from north-east winds.
Which of the statements given above is/are correct?
(a) 1 only
(b) 2 only
(c) Both 1 and 2
(d) Neither 1 nor 2
7. What causes wind to deflect toward the left in the Southern hemisphere?[2010]
(a) Temperature
(b) Magnetic field
(c) Rotation of the earth
(d) Pressure
8. During a thunderstorm, the thunder in the skies is produced by the [2013]
1. Meeting of cumulonimbus clouds in the sky
2. Lightning that separates the nimbus clouds
3. Violent upward movement of air and water particles
Which of the statements given above is/are correct?
(a) 1 only
(b) 2 and 3
(c) 1 and 3
(d) None of the above
9. Consider the following statements about westerlies:[2013]
1. The winds which blow between 30° N and 60° S latitudes throughout the year are known as westerlies.
2. The moist air masses that cause winter rains in North-Western region of India are part of westerlies.
Which of the statements given above is/are correct?

- (a) 1 only
 - (b) 2 only
 - (c) Both 1 and 2
 - (d) Neither 1 nor 2
10. In the South Atlantic and South-Eastern Pacific regions in tropical latitudes, cyclones do not originate. What is the reason? [2015]
- (a) Sea surface temperatures are low
 - (b) Inter-Tropical Convergence Zone seldom occurs
 - (c) Coriolis force is too weak
 - (d) Absence of land in those regions
11. Consider the following statements:[2020]
- 1. Jet streams occur in the Northern Hemisphere only.
 - 2. Only some cyclones develop an eye.
 - 3. The temperature inside the eye of a cyclone is nearly 10°C less than that of the surroundings.
- Which of the statements given above is/are correct?
- (a) 1 only
 - (b) 2 and 3 only
 - (c) 2 only
 - (d) 1 and 3 only
12. Consider the following statements:[2021]
- 1. In the tropical zone, the western sections of the oceans are warmer than the eastern sections owing to the influence of trade winds.
 - 2. In the temperate zone, westerlies make the eastern sections of oceans warmer than the western sections.
- Which of the statements given above is/are correct?
- (a) 1 only
 - (b) 2 only
 - (c) Both 1 and 2
 - (d) Neither 1 nor 2
13. Consider the following statements with reference to the Coriolis force [2024]
- 1. It increases with an increase in wind velocity.
 - 2. It is maximum at the poles and is absent at the equator.
- Which of the statements given above is/are correct?
- (a) 1 only
 - (b) 2 only
 - (c) Both 1 and 2
 - (d) Neither 1 nor 2

PREVIOUS YEAR QUESTIONS - MAINS

- 1. The recent cyclone on the east coast of India was called 'Phailin'. How are tropical cyclones named across the world? Elaborate. [2013]
- 2. What is sea surface temperature rise? How does it affect the formation of tropical cyclones? [2020]
- 3. What is the phenomenon of 'cloudbursts'? Explain. [2024]
- 4. What is the Fujiwhara effect? Explain its impact on the movement and intensity of tropical cyclones? 2026

NCERT Reference

Class 11th - Fundamentals of Physical Geography
Chapter 10: Atmospheric Circulation and Weather System